

KARSAKBAY, KazakhstanWMO: **356620**Lat: **47.833N**Lon: **66.717E**Elev: **1601**StdP: **13.87**Time Zone: **6.00 (E06)**Period: **82-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-17.2	-12.9	-25.0	1.5	-15.5	-20.8	1.9	-10.9	40.1	18.4	32.6	17.9	2.2	270	0.555

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.4	93.1	63.3	89.8	62.4	86.5	61.9	67.5	86.4	65.2	84.6	63.4	82.2	6.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.2	82.5	76.9	57.5	75.0	74.5	55.4	69.4	72.9	32.8	86.2	31.0	84.2	29.6	81.8	90.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.7	20.5	17.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.1	8.6	6.2	18.4	44.5	57.3	68.9	73.1	68.8	56.1	40.9	22.8	13.2		
	DBStd	25.74	10.91	10.54	10.47	10.11	8.01	7.07	5.79	6.39	8.30	8.92	11.00	11.49		
	HDD50	6029	1283	1226	978	218	28	0	0	0	33	306	817	1140		
	HDD65	9696	1748	1646	1443	617	255	40	9	35	284	747	1267	1605		
	CDD50	2414	0	0	0	52	255	566	717	584	216	24	0	0		
	CDD65	604	0	0	0	0	16	156	261	154	17	0	0	0		
	CDH74	7165	0	0	0	14	229	1936	2996	1748	237	5	0	0		
	CDH80	2763	0	0	0	1	28	751	1285	647	51	0	0	0		
	WSAvg	5.4	5.3	5.8	6.1	6.3	6.1	5.4	5.3	4.8	4.4	4.9	5.4	4.5		
	PrecAvg	7.5	0.9	0.4	0.6	0.8	0.6	0.6	0.7	0.4	0.5	0.8	0.6	0.7		
	PrecMax	13.3	3.6	1.2	1.5	3.4	1.4	2.1	3.3	1.3	1.9	2.5	2.2	2.0		
(15) Precipitation	PrecMin	4.3	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	2.2	0.8	0.3	0.4	0.8	0.4	0.5	0.8	0.4	0.6	0.6	0.5	0.5		
	DB	32.4	31.1	48.9	75.9	83.9	97.6	97.5	96.0	86.3	70.6	48.5	39.9			
	MCWB	31.1	29.5	43.3	55.3	60.0	64.7	64.1	63.9	61.7	52.2	42.7	34.1			
	2% DB	29.8	28.3	40.0	69.5	79.6	91.6	93.7	90.7	80.2	65.1	42.8	33.6			
(23) Mean Coincident Wet Bulb Temperatures	MCWB	29.0	26.5	36.9	52.6	57.8	63.6	63.2	62.0	57.6	48.6	37.5	32.3			
	DB	26.8	25.7	35.3	65.1	76.5	87.7	90.6	87.0	76.2	60.6	39.1	31.0			
	MCWB	25.8	24.0	32.8	49.9	57.2	62.7	63.0	60.8	55.5	47.3	35.6	29.3			
	DB	24.4	22.7	32.0	60.9	73.2	84.2	87.2	83.4	72.1	55.3	35.8	28.3			
	MCWB	23.3	21.0	30.0	47.7	55.3	62.2	62.5	59.6	53.3	44.4	33.2	26.5			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	31.9	29.6	43.1	57.5	63.5	81.1	69.9	69.3	64.2	53.9	44.2	34.7			
	MCDB	32.4	30.2	47.4	71.7	77.9	83.2	89.3	89.4	80.7	64.9	48.4	38.3			
	2% WB	29.1	26.6	37.0	53.7	60.3	67.9	67.3	64.3	59.3	50.1	38.8	32.4			
	MCDB	29.7	28.0	39.9	66.1	75.1	86.2	86.6	85.2	76.4	62.3	41.7	34.2			
	5% WB	25.9	24.2	33.0	51.1	58.1	65.0	65.4	62.1	56.6	48.1	36.2	29.5			
(31) Mean Daily Temperature Range	MCDB	26.7	25.7	35.0	63.1	73.6	84.0	84.9	82.4	74.2	58.5	38.9	31.0			
	10% WB	23.2	21.1	30.2	48.8	56.2	62.9	63.7	60.6	54.1	45.5	33.5	26.6			
	MCDB	24.2	22.6	32.0	59.4	71.0	80.8	82.9	80.4	70.9	54.6	36.2	28.1			
	MDBR	12.2	14.5	14.6	19.5	21.8	23.4	22.4	22.7	22.3	19.1	12.3	12.5			
	5% DB	12.2	15.0	15.3	26.9	26.6	27.8	27.5	27.6	28.1	27.2	16.0	14.3			
(36) Clear-Sky Solar Irradiance	MCWBR	11.1	13.1	12.2	13.7	12.9	12.8	11.0	11.6	14.4	14.6	12.7	12.4			
	MCDBR	11.4	14.2	14.5	24.6	23.9	24.6	23.5	24.0	26.8	23.6	13.8	13.8			
	MCWBR	11.1	13.3	12.0	13.3	13.0	13.6	11.2	11.8	15.1	14.8	11.9	12.4			
	taub	0.296	0.329	0.387	0.420	0.428	0.419	0.418	0.393	0.350	0.347	0.329	0.302			
	taud	2.100	2.010	2.046	2.079	2.117	2.209	2.234	2.284	2.359	2.281	2.216	2.145			
(42) All-Sky Solar Radiation	Ebn at Noon	237	254	258	263	265	268	267	269	271	249	224	218			
	Edh at Noon	28	39	44	48	48	44	42	38	32	30	25	24			
(44) All-Sky Solar Radiation	RadAvg	489	847	1221	1653	2094	2217	2114	1924	1495	882	458	351			
	RadStd	80	89	122	151	143	132	104	111	56	79	43	59			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page